

CURRICULAM VITAE

Dr. Atul Dhakar
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Career Objective

To secure a challenging teaching and research position at a reputed institution, enabling the application and growth of technical and academic skills while contributing to organizational development.

Summary of Skills

- **Subject Expertise:** In-depth knowledge of core mechanical engineering concepts, including Engineering Graphics, Basic Mechanical Engineering, Thermodynamics, Fluid Mechanics, Materials Science, etc.
- **Teaching & Instruction:** Experienced in delivering lectures, designing course materials, and fostering an interactive and engaging learning environment for undergraduate and graduate students.
- **Research & Innovation:** Strong research background in fault diagnosis of mechanical systems using signal processing and machine learning techniques, solar water distillation with using PCM material, and MATLAB software. Possesses strong knowledge of writing and publishing SCI and Scopus-indexed papers, with a focus on innovative problem-solving and practical applications.
- **Student Mentorship:** Effective in mentoring students for academic success and professional development, with a focus on critical thinking, creativity, and teamwork.
- **Continuous Learning:** Committed to staying updated with the latest advancements in mechanical engineering through ongoing professional development and participation in academic conferences.

Academic Qualifications

- **Pursuing 2nd M. Tech Degree in Artificial Intelligence & Data Science** from **Indian Institute of Information Technology (IIIT), Ranchi**.
- **Completed Ph.D. Degree in Mechanical Engineering** (August 2025)
Jaypee University of Engineering and Technology (JUET), Guna (M.P.)
Paper Published: 3 SCI | 1 E-SCI | 3 Scopus | 1 Book Chapter | 3 UGC – Approved Journals = 11 Papers.
Research Focus: *Fault Diagnosis of Reciprocating Air Compressor Set-up using Signal Processing and Machine Learning Techniques*. | Under the Supervision of **Dr. Bhagat Singh & Dr. Pankaj Gupta**
- **Completed Master of Technology Degree in Thermal System & Design** (2016)
ShriRam College of Engineering & Management, Gwalior | RGPV, Bhopal | *With Honors*
- **Completed Bachelor of Engineering Degree in Mechanical Engineering** (2010)
Sushila Devi Bansal College of Engineering (KCB Technical Academy), Indore | RGPV, Bhopal | *First Division*
- **Higher Secondary Education with PCM** (2006)
Govt. Excellence School No. 1, Shivpuri | *First Division*

Professional Experience

- **Total Teaching Experience: 13+ Years**

S. No	Institution	Designation	Tenure	Duration
1.	UIT-RGPV, Shivpuri	Assistant Professor	8 Sept. 2025 – Till Now	Working
2.	SKITM, Indore	Assistant Professor	Feb 2021 – July 2024	3 Year, 3 Months
3.	SVCE, Indore	Assistant Professor	Mar 2017 – Dec 2020	3 Year, 9 Months
4.	GICTS Group of College, Gwalior	Assistant Professor	Nov 2014 - Nov 2016	2 Years
5.	AITR, Indore	Lecturer	Aug 2010 – July 2014	4 Years

Key Responsibilities in Current Institute (UIT, RGPV, Shivpuri):

- Assistant Exam Superintendent (RGPV Exam).
- Coordinate Department Level Activities
 - Research & Innovation Activities
 - Conduction of Workshop / Seminar / Conferences
 - Students Visit Outside & Inside the Institute
- Time – Table Coordinator (Department Level)
- Training & Placement Department Representative
- Lab In charge: I.C. Engine, BME & Engineering Drawing
- Subject Taught: Engg. Graphics, TOM, SOM and Mechanical Vibration.

Key Responsibilities in Previous Institutes:

- Held the position of Counseling Coordinator in the **Counseling Department for 5 years** at SVCE & SKITM College, Indore, overseeing and managing student counseling related work such as students counselling, MPDTE registration and reporting process, etc.
- Executed managerial and administrative tasks, including leadership roles.
- Organized expert lectures, workshops, and industrial visits for students and faculty.
- Served as Faculty Advisor for **Eco-Kart Race** (2018, 2019) at SVCE, Indore.
- Arranged career counseling visits for school students to the college campus.
- Mentored assistant teachers and new hires to enhance their work efficiency.
- Guided students on career development and explored diverse career opportunities.
- Prepared teaching materials and utilizing them to support the progress of students.

Subjects Taught

- Engineering Graphics (ED), Basic Mechanical Engineering (BME), Thermodynamics, Heat and Mass Transfer (HMT), Fluid Mechanics (FM), and Material Technology, etc.
- Good Knowledge of **MATLAB & WEKA** Software for research purposes.

Paper Published, Paper Reviewed, Patents and Membership

- **ORCID:** 0000-0003-3457-4662 | **Scopus ID:** 58866418000 | **Google Scholar Profile:** Citations: 39
- **Ph.D. Thesis:** Available on Shodhganga (Link): <http://hdl.handle.net/10603/659555>

SCI Indexed Journals (3 Papers Published)

- Atul Dhakar, Bhagat Singh and Pankaj Gupta, Fault Identification of Reciprocating Air Compressor using Signal Processing Techniques and Kurtosis Index-Based Bubble Cloud Analysis, Journal of the Chinese Institute of Engineers, 2025. <https://doi.org/10.1080/02533839.2025.2471376>. (**Taylor & Francis**)
- Atul Dhakar, Bhagat Singh, and Pankaj Gupta, Diagnosing Faults in Rolling Bearings of An Air Compressor Set-up using Local Mean Decomposition and Support Vector Machine Algorithm, Journal of Vibration Engineering & Technologies, 2024. <https://doi.org/10.1007/s42417-024-01275-6>. (**Springer**)
- Atul Dhakar, Bhagat Singh, and Pankaj Gupta, Fault Diagnosis of Air Compressor Set-Up using Decision Tree based J48 Classification Algorithm, Journal of Engineering Research, 2023. <https://doi.org/10.1016/j.jer.2023.09.028>. (**Elsevier**)

Scopus and E-SCI Indexed Journals (4 Papers Published)

- Atul Dhakar, Bhagat Singh, and Pankaj Gupta, Fault Diagnosis in Air Compressor System using Advanced Signal Processing and Discriminant Analysis-based (LDA and QDA) Machine Learning Techniques, Noise & Vibration Worldwide - Accepted. (**Sage Journal**)
- Atul Dhakar, Bhagat Singh, and Pankaj Gupta, Comparative Performance Analysis of Different Types of K-Nearest Neighbor (K-NN) Classifiers for Fault Diagnosis of Air Compressor Setup, Engineering Research Express, 2024. <https://doi.org/10.1088/2631-8695/ad5497>. (**IOP Science**)
- Atul Dhakar, Bhagat Singh and Pankaj Gupta, Fault Diagnosis of Air Compressor System using Local Mean Decomposition (LMD) and Logistic Regression (LR) Machine Learning Classifier, International Conference on Mechanical and Materials Engineering. (**STM Journal**)
- Atul Dhakar, et al., Mechanical characterization of FRBC prepared from coir fibers, International Journal of Engineering and Advanced Technology, 2019. <https://doi.org/10.35940/ijeat.B3926.129219>.

Book Chapter/Lecture Notes (Scopus) – (1 Published)

- Atul Dhakar, Bhagat Singh, and Pankaj Gupta, Diagnosing Faults of Reciprocating Air Compressor (RAC) Setup Using Signal Processing Technique and Machine Learning Approach, Recent Advances in Mechanical Engineering. Lecture Notes in Mechanical Engineering, 2024. https://doi.org/10.1007/978-981-97-4947-8_37. (Springer)

UGC Approved Journals

- Atul Dhakar and Neelesh Soni, Performance improvement of single slope solar water distillation plant by using paraffin wax martial, International Journal of Recent Scientific Research, 2016.
- Atul Dhakar and Neelesh Soni, A review of design and fabrication of single slope solar water distillation system using phase change material, Journal of Advances in Science and Technology, 2015.
- Atul Dhakar, Survey and Case Study of Scheme for Installing Solar Power Plants Instead of Waiving Electricity Bill for Farmers, International Research Journal of Engineering and Technology, 2020.

Paper Reviewed: Various SCI and Scopus Indexed Journals

- The Journal of Supercomputing (SCI Indexed)
- Measurement Science and Technology (SCI Indexed)
- Engineering Research Express (Scopus & E-SCI Indexed)
- Journal of Brazilian Society of Mechanical Sciences and Engineering (SCI Indexed) – 3 Paper
- Journal of Physics (SCI Indexed).

Design Patents Published

Application Nos:

- 484785-001 – Under Process (2026)
- 469219-001 - Applied (Under Process) - 2026
- 469221-001 – Serving Robot – Published (2025)
- 469222-001 – Sprayer Robot – Published (2025)
- 469218-001 - Carrier Lifting Robot – Published (2025)
- 469220-001 – Fire Fighting Robot – Published (2025)

Professional Memberships

- Professional Member, Ignited Minds Professional & Academic Research Consortium (IMPARC).
- Permanent Member, International Association of Engineers (Membership Number: 246598).
- Lifetime Member, Vigyan Bharti.

Offline/ Online Certificate Courses and FDPs

- **International Conference on Mechanical and Materials Engineering (ICMME-2025) - Offline**
 - **Organized by:** Jaypee University, Guna (M.P.) | Presented Research Paper | April 28-30, 2025
- **Completed Online Diploma in Electric Vehicle Technology**
 - Organized by: Alison (ID – 45855047) | FAS – 95% | Hours Completed – 28h.
- **Offline Advanced Certification: Solar Energy Technology and Management (6 Months)**
 - Organized by: Madhav Institute of Technology & Science, Gwalior.
- **Offline Short-Term Course: Research Perspective in IC Engines (1 Week)**
Organized by: Govt. College of Engineering Amravati | Under: AICTE-QIP.
- **NPTEL Course: Fundamentals of Manufacturing Processes (14 Weeks)**
 - Organized by: IIT Roorkee | Under: MHRD, Govt. of India.
- **NPTEL FDP: Fundamentals of Manufacturing Processes (7 Weeks)**
 - Organized by: IIT Roorkee | Under: MHRD, Govt. of India.
- **Offline Training Program: Master CAM (1 Week)**
 - Organized by: IGTR, Indore | Under: MSME Govt. of India.
- **Online Certification: LaTeX Software (2 Months)**
 - Organized by: IIT Bombay.
- **National E-Conference: Recent Trends in Fluid Dynamics Research (3 Days)**
 - Organized by: NIT Rourkela.
- **Online Certification: Introduction to Thermodynamics (8 Weeks)**
 - Organized by: University of Michigan, Ann Arbor (Coursera).

- **International FDP:** Research & Development in Material Behavior, Processing, and Characterization Techniques (1 Week)
 - Organized by: GLA University, Mathura | Under: IIM Mathura and Panjab University.
- **FDP:** Recent Advances in Mechanical Engineering (1 Week)
 - Organized by: SGSITS, Indore | Under: TEQUIP-III.
- **FDP:** Innovation, Entrepreneurship, and its Relevance in Industry 4.0 (1 Week)
 - Organized by: TERNA Engineering College, Mumbai | Under: Indian Institution of Industrial Engineering.
- **FDP:** Robotics and Automation (5 Days)
 - Organized by: SAGE University, Indore | Under: IIITDM Jabalpur.

Participated in Webinars and Workshops

- **Webinar Series:** CAE Driven Product Design (5 Days)
 - Organized by: Dept. of Automobile Engineering, RIT, Rajaram Nagar | Under: Altair Engg. Pvt. Ltd & SAENIS Effi-Cycle 2020.
- Participated in the National Webinar on "Writing a Research Paper & Publishability: Issues & Concerns" organized by the IQAC of Vanita Vishram Women's College, Surat.
- Completed a One-Day Online Certification Course on "10 Reasons to Get and Use an ORCID ID" conducted by Researcher Academy, Elsevier.
- Attended a One-Day Webinar on "Intellectual Property Awareness Program" held at SAGE University, Indore, M.P., on June 29, 2019.

Workshops & Expert Talk Organized as a Coordinator

- **One Day Workshop:** Patent Filing Process
 - Speaker: Mr. Nitesh Pandey (Aisa & India Book Record Holder – Filed 75 patent in a Single Day)
 - Organized at: UIT, RGPV, Shivpuri.
- **Two-Day Online Workshop:** Energy Audit
 - Sponsored by: RGPV, Bhopal (Under TEQIP-III)
 - Organized at: Swami Vivekanand College of Engineering, Indore (June 7-8, 2020).
- **Three-Day Offline Workshop:** Engine Assembly
 - Organized in collaboration with: Laxmi Motors, Indore
 - Organized at: SVCE, Indore.
- **Expert Talk:** Fundamentals of Psychometric and Recent Trends in RAC
 - Speaker: Dr. Neelesh Soni (Ph.D., IIT Kharagpur)
 - Organized at: SVCE, Indore.
- **Expert Talk:** Recent Trends in Mechatronics
 - Speaker: Mr. Bhupendra Singh (Robotronix Eng. Tech. Pvt. Ltd., Indore)
 - Organized at: SVCE, Indore.
- **Expert Talk:** Empowering Engineering Through Research
 - Speakers: Dr. Amit Sharma, Dr. Manish Patidar, and Dr. Abhishak Verma (Jaypee University of Engineering and Technology, Guna)
 - Organized at: SKITM College, Indore (March 4, 2024).

Industrial Visit Organized as a Coordinator

- Successfully organized industrial visits for B. Tech students to the following companies: Auto Cluster, Pithampur, NATRAX, Pithampur, Dainik Bhaskar KBA Printing Plant, Indore, INDO German Tool Room, Indore, Mahindra Two Wheelers Ltd., Pithampur and Jash Engineering Ltd., Indore.

Academic Achievements

- Achieved the highest SGPA in M. Tech for 3rd and 4th semesters (82%, 90%).
- Designed multiple banners for college promotion activities at SVCE, Indore, over a span of 2 years.
- Designed and created the official college brochure and banners for SKITM College, Indore.
- Managed YouTube Channel: The College Coach (24×7) (750+ videos, 18,000+ subscribers).

Personal Information

- Date of Birth: 25/11/1988
- Father Name: Mr. S.S. Dhakar
- Mother Name: Mrs. Vimla Dhakar
- Nationality: Indian
- Language Proficiency: Hindi & English
- Permanent Address: 87, Gandhi Colony, Shivpuri (M.P.) Pin - 473551

Declaration

I hereby declare that the above-mentioned information is correct up to my knowledge and I bear the responsibility for the correctness of the above-mentioned particulars.

Date:

Signature

Place: